RESEARCH

Deep Learning Integrated Multi-Cloud Healthcare System for Intelligent Heart Disease Risk Prediction

S. V. Anandhi · Mohana Suganthi · B. Ramakantha Reddy · S. Priya

Received: 14 August 2025 / Revised: 18 February 2026 / Accepted: 21 May 2026
© The Author(s) 2026

Abstract
Heart disease (HD) prediction has become an essential focus in healthcare, as it holds the potential to significantly reduce the global burden of cardiovascular diseases by enabling early detection and prevention. HD is a critical area of medical research aimed at reducing mortality rates by enabling early diagnosis and timely intervention. In this work, a novel multi-Cloud-based Unified Risk Evaluation for Heart Disease (CURE-HD) has been proposed which uses hybrid deep learning model for risk prediction in smart healthcare environments. Initially, data acquisition was performed using IoT devices and pre-processing at the Distributed Edge Server (DES). The pre-processed data is then fed into an Attention-based Long Short-Term Memory-Capsule Network (ALSTM-CapsNet) for classifying patient risk into three levels: low, moderate, and high based on Electronic Health Records (EHRs) and current sensor data. The risk classification results are managed by a Distributed Reinforced Scheduler (DRS) for Heart Disease Crisis Planning (HDCP), where patient data is scheduled according to their risk level using Multi-Agent Deep Reinforcement Learning (MA-DRL). Subsequently, the scheduled reports are sent to the Multi-Cloud Server (MCS) for heart disease detection, utilizing tailored AI models: Support Vector Machine (SVM) for low-risk, Convolutional Neural Network (CNN) for moderate risk, and Transformer based CNN (T-CNN) for high-risk patients. From the experimental results, the proposed CURE-HD demonstrated superior performance by achieving an accuracy of 99.01%. From the results, the proposed CURE-HD increases the overall accuracy by 11.38%, 10.92%, and 5.51% for FRIEND, DEEP-CARDIO, and Heart Sense respectively.

Keywords Heart diseases · Deep learning · Distributed reinforced scheduler · Multi-Cloud Server

1 Introduction

The Internet of Things (IoT) technology as an emergency has heralded a new era in the healthcare, which is simply varying the terrain of predicting and managing heart diseases. This techno-revolution imparts linked machines and indicators to acquire and analyze physiological information in real-time [1], which has the capabilities to provide unrestricted opportunities to early diagnosis and continuous surveillance of cardiovascular wellbeing. Potential of manging and tracking heart disease have suggestively been improved by integrating the IoT technology in healthcare system [2] that has resulted in better patient care and clinical outcomes. The greatest

https://doi.org/10.1007/s44196-026-01405-w

Published online: 11 July 2026

reflective impact of the IoT technology in the management of heart diseases is the ability to enable real-time monitoring of patients [3]. Conventional style of cardiovascular health observations obviously involves patient that is traditionally necessitated to visit healthcare centers frequently which may not only be troublesome but also resource draining [4]. The IoT solution can solve the identified challenge by allowing the collection and the transmission of physiological data taken continuously and almost immediately in the real-life situation of the patients. The devices [5] are IoT-enabled as smart monitors and wearable sensors, which simultaneously detect vital health indicators, such as electrocardiograms (ECG), blood pressure and heart rate.

The measurements of ECG signal are particularly crucial in identifying ischemic events, arrhythmias and other abnormalities of the heart [6]. These readings allow a describe version of the electrical activity of the heart allowing the early identification of anomalies that may indicate competent heart disorders. Blood pressure measurement is necessary in the determination of hypertension and other conditions associated with it [7], in cases where the heart rate would help in the estimation of the overall cardiovascular functioning and that can bring forth the pattern of revealing of heart diseases [8]. Progressive machine learning (ML) methods take into consideration the real-time nature of this data collection and process and analyse the received data. The algorithms are to determine the very minor curves and fluctuations which might indicate the onset of HD within data sets of ECG large volumes [9] an example is the usage of the ML models to identify irregularities that may otherwise go undetected in the course of the sporadic numerical calculations.

The IoT can be used in the healthcare sector to monitor the health status of all patients, including blood pressure and ECG, to detect heart disease early and intervene in time [10]. This real time data helps medical practitioners to develop individualized therapy, minimize complications and unwarranted hospital admissions [11]. It also helps in increasing the use of resources and advancing of cost-effective care and patient outcomes. Remote monitoring helps in benefiting the patients by reducing the frequency of clinical visits and promoting active health management [12].

Despite these benefits, several challenges remain. IoT devices may suffer from noise, inaccuracies, or sensor failures, which can lead to unreliable diagnostic outcomes. Thus, regular calibration and maintenance are essential to ensure data quality [13]. In addition, effective task scheduling in IoT environments remains a major challenge, as delays or missed alerts can directly impact patient safety, requiring intelligent real-time scheduling mechanisms [14]. Machine learning models may also exhibit performance bias if the training data is not diverse or representative. Improving fairness, transparency, and interpretability of such models is essential for clinical adoption [15]. To address these limitations, this paper proposes a *novel* multi-Cloud-based Unified Risk Evaluation for Heart Disease (CURE-HD) model that performs accurate multi-level risk classification and incorporates an intelligent scheduling mechanism to enhance diagnostic precision and healthcare response. The novelty and contributions of the CURE-HD model are articulated as follows:

- The main objective of the proposed CURE-HD model is to build a comprehensive heart disease risk prediction framework that effectively integrates EHR data with real-time IoT sensor information to classify patients into low, moderate, and high-risk categories, enabling accurate and reliable risk assessment.
- The proposed model introduces a novel dynamic scheduling mechanism based on Multi-Agent Deep Reinforcement Learning (MA-DRL), which efficiently allocates healthcare resources by considering patient risk scores, waiting times, and resource availability, thereby improving response efficiency and workflow management.
- The CURE-HD model provides risk-specific health recommendations generated from the AI-based classification outputs, offering personalized and actionable guidance that enhances patient care and supports timely clinical decision-making.
- The efficiency of the developed CURE-HD technique has been assessed using specific metrics such as accuracy, precision, recall and f1-score.

The rest of this research is explained as follows: Sect. 2 summarizes the recent literature of heart disease detection using IoT sensors. The proposed CURE-HD is explained in extensive detail in Sect. 3. Section 4 is the result of the analysis and discussion whereas Sect. 5 explain conclusion and future work.

2 Literature Review

Many researchers have used different methods to develop an IoT-based HD prediction system. Among those, some of the techniques have been discussed in this section.

In 2023, Pati et al. [16] introduced a Fog-empowered framework using the ENsemble Deep learning (FRIEND) model in order to carry out real-time analysis in cardiac patients, which makes use of a combined dataset to diagnose heart disease. The experiments' accuracy, True Positive Rate (TPR), True Negative Rate (TNR), and F1 scores were 94.27%, 96.09%, 75.44%, and 96.83%, respectively, when the deep neural network was used in conjunction with the bagging and hard-voting classifiers.

In 2023, Raheja et al. [17] introduced a cloud-based, IoT-enabled approach for remote ECG monitoring that could save lives in rural areas, especially during golden hour. The suggested model classified cardiovascular illness using a convolutional neural network (CNN). The CNN classifier has an average accuracy of 99.12% in classifying heartbeats into five different types of arrhythmias; in contrast, the proposed model's sensitivity and specificity are 100% and 99.9%, respectively.

In 2024, Yashudas et al. [18] introduced a new recommendation system (DEEP-CARDIO) to provide early diagnosis, treatment, and dietary advice for cardiovascular illnesses. Through a mobile application, the recommendation system gives cardiac patients food and activity advice based on the identified data. While the MABC-SVM, HCBDA, and MLbPM approaches reach 86.91%, 88.65%, and 93.63% overall accuracy, respectively, the suggested DEEP CARDIO method achieves a high precision.

In 2024, Durga et al. [19] introduced a cardiac care strategy that uses edge cloud computing and IoT technology to help decision-making and offer quick, automated responses. In order to forecast patient situations and provide easily available healthcare services, the system links smart devices, sensors, and healthcare providers. Experimental results show that the smart cardiac system outperformed conventional techniques like k-nearest neighbors, support vector machines, and discrete wavelet-based convolutional neural networks, achieving 99.3% accuracy with decreased network delay and response time.

In 2024, Ramesh, B and Lakshmanna, K [20] introduced an ideal Scrutiny Boosted Graph Convolutional LSTM (O-SBGC-LSTM), which is a modified SBGC-LSTM enhanced by the Eurygaster Optimization Algorithm (EOA) to fine-tune hyperparameters for predicting diabetic conditions and facilitating early intervention. Most studies indicated that O-SBGC-LSTM achieved impressive results overall, boasting an accuracy exceeding 98%. The research demonstrated that the introduced hybrid deep learning model surpassed conventional machine learning methods in almost every test that evaluated its effectiveness.

In 2024, Mandava, M [21]. introduced a hybrid DLintelligent system for predicting cardiovascular disease. To predict cardiovascular disease, the suggested model included an improved deep residual shrinkage network (IDRSNet), which is based on deep learning. According to the test results, the hybrid deep learning system that was suggested has exceeded the existing methods in terms of the accuracy of CVD prediction.

In 2025, Ghilzai et al. [22] introduced the Heart Sense method, which predicts heart diseases more accurately and precisely than current methods, and provides patients and medical professionals with timely notifications so they can respond to the situation as effectively as possible. The suggested model outperformed the current methods for predicting heart disease.

In 2025, Sekar, J. and Aruchamy, P [23]., introduced an innovative combined optimization method aimed at enhancing classification precision and reducing computational duration in intelligent healthcare applications. The best features are identified using the combined Arithmetic Optimization and Inter-Twinned Mutation-Based Harris Hawk Optimization (AITH2O) technique. The results of the simulation indicate that the suggested hybrid classifier achieves an impressive accuracy rate of 99. 28% and a true positive rate of 99. 46% when compared to current leading-edge methods.

In 2025, Chichani, Y.S. and Kasar, S.L [24]. introduced a heart disease prediction model that integrates IoT technology and applies a modified BiLSTM classifier based on Finch hunt optimization (FHO modified

BiLSTM). This method not only boosts the accuracy of predictions but also makes it possible to detect complex temporal patterns in the patient's data via predictive modelling. The testing results showed that the suggested FHO-modified Bi-LSTM achieved the following F1-score, precision, recall, and accuracy values: 95. 17%, 96. 52%, 93. 86%, and 97. 24% respectively, with 80% of the data being allocated for training. Thus it surpassed all other established methods.

In 2025, Tarek et al. [25] introduced CVD-SO, a highly innovative framework that employs snake optimization (SO) for the analysis of CVD data. The combination of ML with the SO algorithm into a single framework led to the creation of an extremely accurate CVD diagnostic model. If this paradigm is adopted along with CVD actively combating through early interventions, it will not only benefit healthcare systems by reducing mortality rates but also significantly decrease the wide burden of illnesses linked to CVD and the related death rates.

Table 1 illustrates the Comparative Analysis of existing works. Despite notable advancements in IoT-enabled cardiovascular prediction systems, significant research gaps remain in existing studies. Most prior works rely on single-modality data—such as ECG images, isolated sensor signals, or EHR values—which limits their ability to perform holistic and clinically meaningful assessment. Additionally, current models largely focus on binary or coarse prediction and do not offer multi-level (low, moderate, high) risk stratification required for effective triage. Although IoT environments produce continuous high-volume data streams, existing frameworks lack intelligent scheduling or resource prioritization mechanisms, resulting in increased latency and reduced scalability. Furthermore, deep learning models used in earlier studies often fail to effectively fuse temporal EHR patterns with spatial physiological signals, leading to suboptimal diagnostic accuracy. To address these limitations, this study proposes the CURE-HD framework, which integrates multi-modal data using a hybrid ALSTM-CapsNet architecture, performs detailed multi-level risk classification, incorporates a Multi-Agent Deep Reinforcement Learning–based scheduling mechanism for optimized patient handling, and leverages a multi-cloud environment for scalable and real-time heart disease prediction.

Table 1 Comparative Analysis of existing works

Authors, Year & [Roll No.]	Techniques Used	Dataset	% Accuracy	Major Contributions	Limitations
Pati et al. 2023 [16]	FRIEND Framework (Fog + Ensemble Deep Learning)	Combined IoT + EHR dataset	94.27%	Real-time fog–cloud diagnosis using ensemble classifiers.	Lower accuracy and not scalable for large IoT streams.
Raheja et al. 2023 [17]	Deep Learning CNN	ECG-based arrhythmia dataset	99.12%	Reliable heartbeat classification with strong sensitivity.	ECG-only, lacks multi-sensor integration.
Yashudas et al. 2024 [18]	DEEP-CARDIO Recommendation System	IoT cardiac dataset	97.35%	Early prediction with diet/activity recommendations.	No multi-risk stratification or cloud scheduling.
Durga et al. 2024 [19]	SmartCardio (Edge–Cloud Intelligence)	IoT + Edge Cloud dataset	98.30%	Low-latency risk prediction via edge computing.	Limited hybrid DL integration.
Ramesh & Lakshmanna 2024 [20]	O-SBGC-LSTM + EOA Optimization	Diabetic/CVD dataset	98.00%	Improves early detection using hybrid DL + optimization.	Computationally expensive.
Mandava 2024 [21]	IDRSNet-Based Hybrid DL	CVD dataset	97.80%	Improved CVD prediction using residual shrinkage networks.	Accuracy not clearly benchmarked; single-source data.
Ghilzai et al. 2025 [22]	Heart Sense (IoT + Deep Learning, ECG Images)	ECG Image dataset	98.89%	Strong ECG-image classification with IoT integration.	Works only on images; no EHR fusion.
Sekar & Aruchamy 2025 [23]	Hybrid AITH2O + SANFIS	Smart Health dataset	98.70%	Optimal features via hybrid optimization improves prediction.	High computational overhead.
Chichani & Kasar 2025 [24]	FHO-Optimized Bi-LSTM	IoT-enabled EHR dataset	97.24%	Extracts temporal patterns more effectively.	Lower performance than latest DL models.
Tarek et al. 2025 [25]	Snake Optimization-Based ML Framework	CVD dataset	97.50%	Rapid feature selection for early CVD detection.	Does not incorporate deep learning; lacks multimodal fusion.

3 CURE-HD Model

In this section, a novel CURE-HD technique has been proposed. Figure 1 shows the overall block diagram of the proposed CURE-HD model. Initially, the data such as EHR data and sensor data has been collected using IoT devices. The EHR data has been pre-processed using data transformation and missing data handling and then the sensor data has been pre-processed using data filtering and normalization. The pre-processed data is then fed into an Attention-based Long Short-Term Memory-Capsule Network (ALSTM-CapsNet) for classifying patient risk into three levels: low, moderate, and high based on Electronic Health Records (EHRs) and current sensor data. The risk classification results are managed by a Distributed Reinforced Scheduler (DRS) for Heart Disease Crisis Planning (HDCP), where patient data is scheduled according to their risk level using Multi-Agent Deep Reinforcement Learning (MA-DRL). Subsequently, the scheduled reports are sent to the Multi-Cloud Server (MCS) system for efficient prediction.

3.1 Data Collection

The proposed heart disease prediction model gathers data from two main sources. The initial source has been composed of the physiological data of the patients which includes the various sensors and measurements taking place at different locations. Electroencephalogram (EEG), Electromyogram (EMG), Electrocardiogram (ECG), cholesterol, blood oxygen, and glucose levels, heart rate, and blood pressure are among the measurements taken. The data is collected through IoT devices and sent to the DES for pre-processing.

Figure 2 depicts the different physiological signals and biomarkers that are monitored in the human body, such as EEG, ECG, and EMG signals, as well as heart rate, respiration rate, blood pressure, oxygen level, cholesterol level, and blood glucose level. These signals and biomarkers are of paramount importance for a proper and

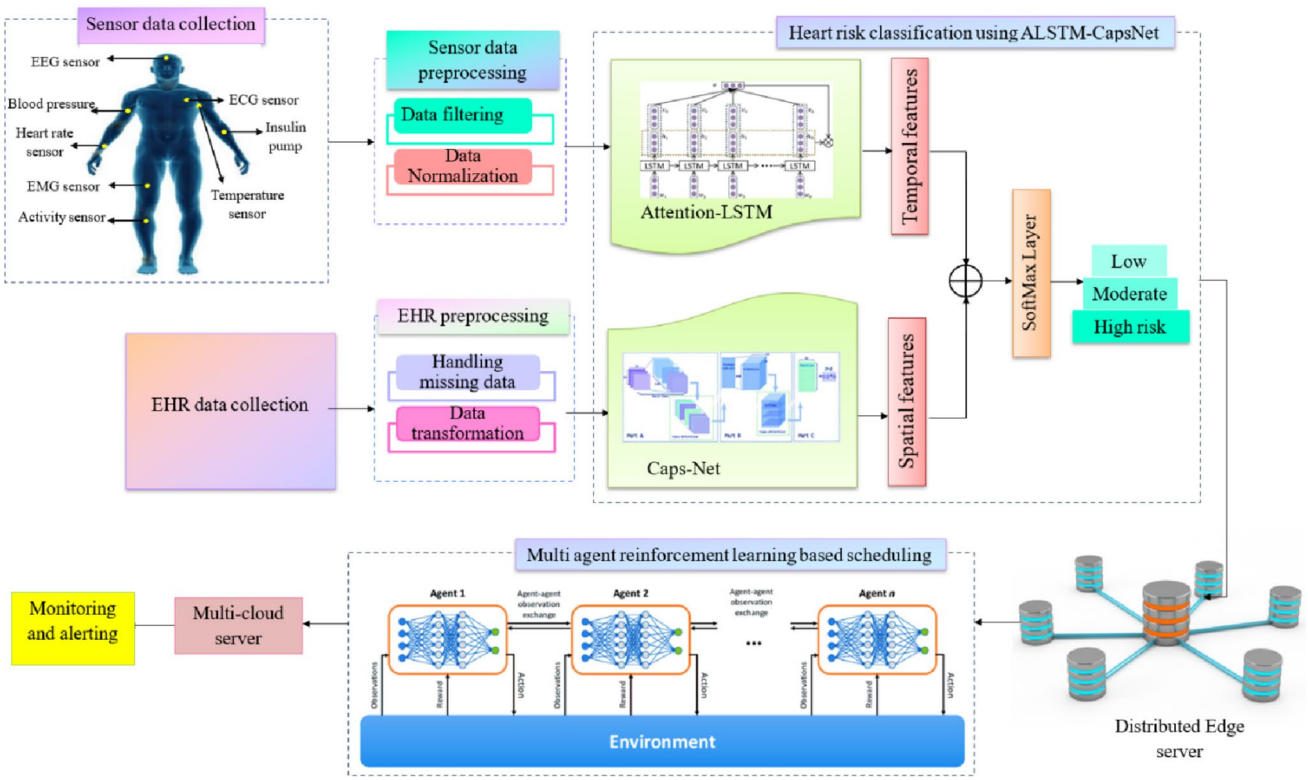


Fig. 1 CURE-HD model

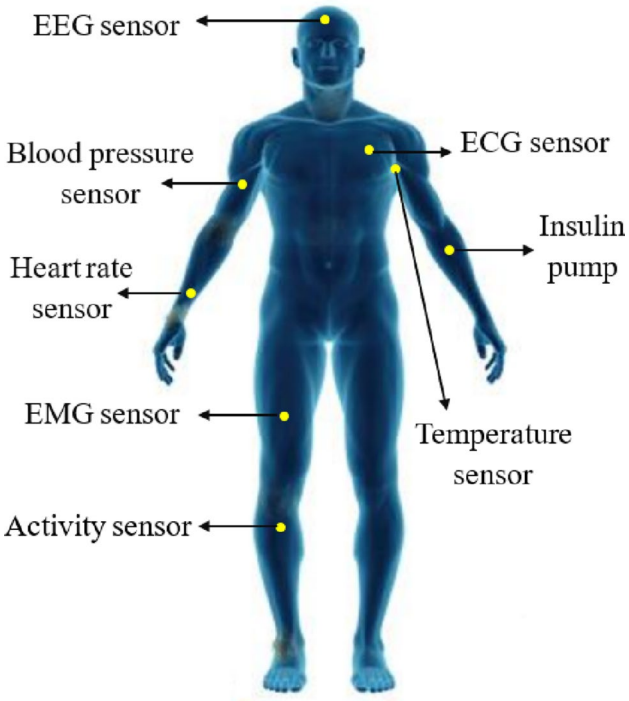


Fig. 2 Multi-Physiological sensor-based signal gathering in a human body for Heart Disease Diagnosis

comprehensive health monitoring system to be in place for heart disease diagnostics. EHR, on the other hand, is used as a secondary data source that contains the medical histories of the patients, laboratory reports, and observation records, all of which are backed up in a cloud database.

3.2 Preprocessing

Preprocessing is the process of eliminating unnecessary information from the input data and transforming it into a format suitable for deep learning analysis. Since the proposed CURE-HD framework uses two heterogeneous data sources namely IoT sensor data and Electronic Health Records (EHR), dedicated preprocessing is applied to each modality to ensure noise-free, normalized, and consistent input before feeding it into the classification model.

3.2.1 Sensor Data Preprocessing

The heart status data $\mathcal{H}_{\mathscr{a}}$ collected from sensors often contains noise and may be unevenly distributed. To address this, we apply a Median Filter (MF) for noise reduction and use Min-Max normalization for data scaling.

(i) Data Filtering

Initially, the data are acquired $\mathcal{H}_{\mathscr{a}}$ includes environmental factors, patient movement, and sensor impairments. After that, the data undergoes an MF process to forecast the future states on the basis of the past data. The MF, which is a non-linear digital filtering technique, is very efficient in eliminating the noise while what's crucial in the data is still very well drawn out. For the used data set $[\mathcal{H}_{\mathscr{A}}]_{\mathbb{i}-1}^{\mathbb{N}}$, where $\mathbb{N}$ represents the total number of data points, the median filter can be mathematically expressed in Eq. (1):

$$\mathcal{Z}_{\mathbb{i}} = MeDian\left\{\mathcal{H}_{\mathscr{A}-\mathscr{k}}, \ldots \ldots \mathcal{H}_{\mathscr{A}}, \ldots ..\mathcal{H}_{\mathscr{A}+\mathscr{k}}\right\} \tag{1}$$

In this expression, $\mathcal{Z}_{\mathbb{i}}$ denotes the filtered output at position $\mathbb{i}$, and $\mathscr{k}$ specifies the size of the filtering window. The window size of $2\mathscr{k}+1$ confirms that the median is computed by including $\mathscr{k}$ neighboring points on either

side of $\mathcal{H}_{\mathscr{A}}$. To assess the performance of the filtering, the Signal-to-Noise Ratio (SNR) can be computed as follows in Eq. (2):

$$SNR = 10\log_{10}\left(\frac{\sum_{\mathrm{i}-1}^{\mathcal{N}}\left(\mathcal{H}_{\mathscr{A}} - \check{\mathcal{H}}_{\mathscr{A}}\right)^2}{\mathcal{N}\rho^2}\right) \tag{2}$$

(ii) Data Normalization

The min-max normalization method adjusts feature values to a predetermined range, classically between 0 and 1, which improves the performance of the proposed model. For an original feature value $\mathcal{H}_{\mathscr{A}NorM}$, the min-max normalization can be described using the following Eq. (3):

$$\mathfrak{X}_{NorM} = \frac{\mathfrak{X}_{\sigma}^{\rho} - \mathfrak{X}_{Min}}{\mathfrak{X}_{Max} - \mathfrak{X}_{Min}} \tag{3}$$

In this context, $\mathfrak{X}_{NorM}$ represents the normalized value, $\mathfrak{X}_{Min}$ is the least value of the feature, and $\mathfrak{X}_{Max}$ is the extreme value of the feature. Where, the filtered data point is denoted the maximum value of the feature. Additionally, the filtered data point is denoted as $\mathfrak{X}_{\sigma}^{\rho}$ denotes the mean and standard deviation of the filtered data.

3.2.2 EHR Data Preprocessing

The EHR data often contains noise and redundancies. To address this, preprocessing techniques such as handling missing values and data transformation has been performed.

(i) Handling Missing Values

Missing values must be handled correctly so that ML models may make accurate predictions using all available data. Missing values are removed by eliminating rows or columns containing null values. A column can be completely removed if more than half of its rows are null.

(ii) Data Transformation

Data transformation converts raw numeric data into a structured format by normalizing values, scaling metrics, and ensuring consistency. This makes the data easier to use and raises the efficacy of analysis and deep learning models in predicting the heart risk.

3.3 Heart Risk Classification Using ALSTM-CapsNet

The sensor and EHR data that have been preprocessed are fed into the proposed ALSTM-CapsNet model to classify heart risks. The structure is able to integrate the strengths of Attention-based LSTM on structured EHR data and Capsule Networks (CapsNet) on multi-physiological IoT sensor signals. This two-branch architecture retrieves complementary temporal and spatial-temporal patterns, which allows the model to retrieve complex, multidimensional patterns of patients. Figure 3 shows that the ALSTM-CapsNet is composed of two feature extractors levels, which include EHR and IoT sensor stream layers. These features are extracted and then the features are combined to group the patients into three risk groups low, moderate and high. The CURE-HD system will be aimed at integrating two types of vital information, long-term medical records and real-time physiological signals. Attention-based LSTM (ALSTM) module works on the Electronic Health Record (EHR) data and determines the most significant clinical attributes using the attention system, where the model is selected to pay attention to the medically significant parameters. Meanwhile, the Capsule Network (CapsNet) processes the signals of the IoT sensor, including ECG, blood pressure, heart rate, and other physiological patterns. Capsules are used to encapsulate the model with spatial and structural variation of these signals better than traditional CNNs. The ALSTM-CapsNet model has been used to classify the heart risk as the pre-processed sensor data and EHR data is the input. The ALSTM-CapsNet model is the integration of EHR and IoT sensor data in extracting multiple

 https://doi.org/10.1007/s44196-026-01405-w

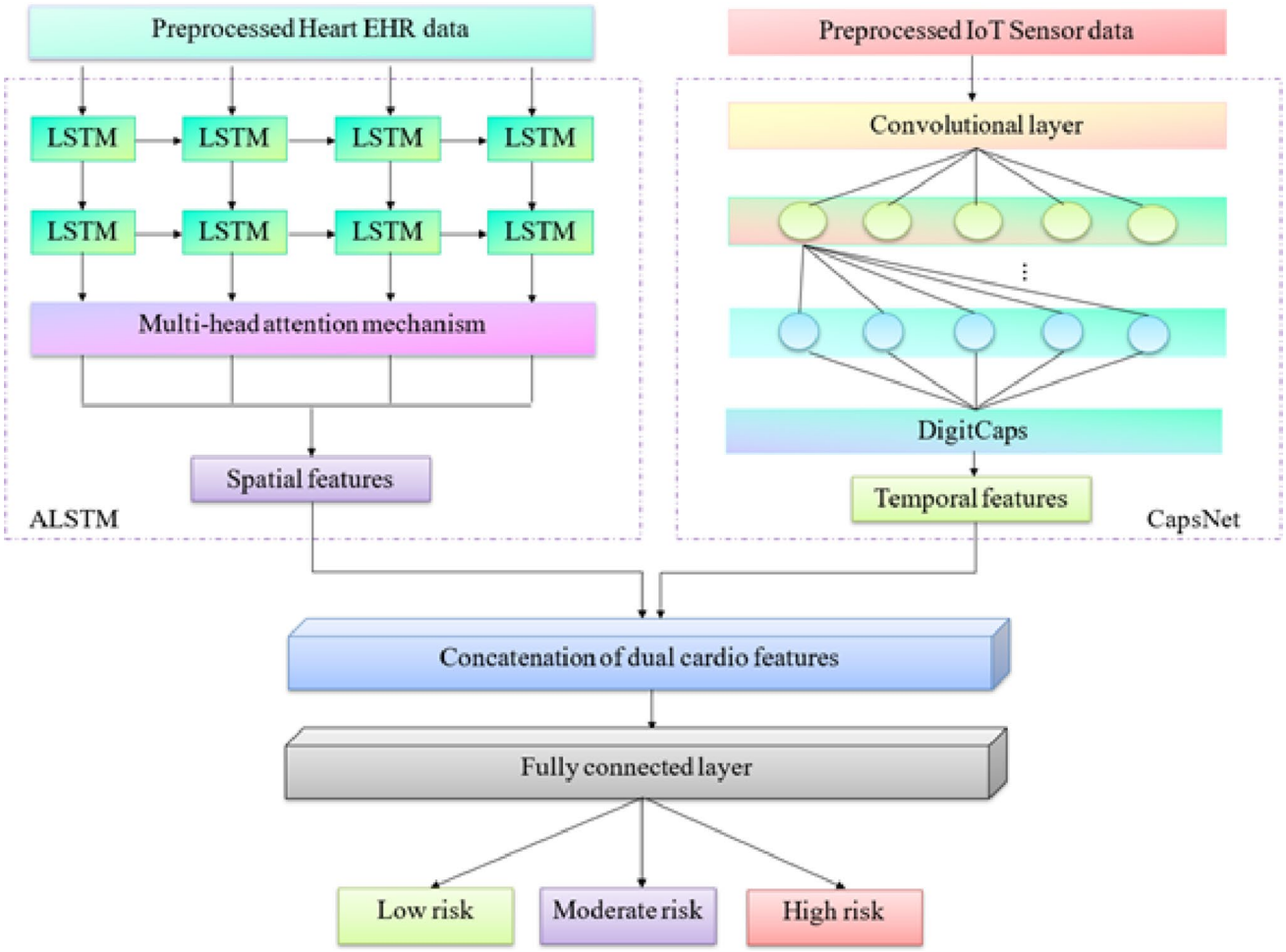


Fig. 3 ALSTM-CapsNet architecture

features that classify risk. The idea here is to find multidimensional characteristics of time series data and this goes a long way in improving the accuracy of prediction. The ALSTM-CapsNet is an architecture with 2 layers whereby the first layer is the EHR feature extraction layer, the second layer is the IoT sensor feature extraction layer, as illustrated in Fig. 3 which represents the risk classification DL model.

3.3.1 Attention based LSTM for EHR Data Feature Extraction

The EHR data feature extraction layer is developed based on the conglomeration of several components. It involves two-layer attention network and two different attention mechanisms. This method entails two-layered stacked LSTM network which employs simplified and context-dependent attention method. Also, the layer of feature extraction combined with LSTM networks allows effectively capturing complex associations between various health measurements and patient records and creates more descriptive and informative feature representations.

The vanishing gradient problem that is present in the traditional Recurrent Neural Networks (RNNs) is resolved by the LSTM unit using its sophisticated structure. It takes three kinds of gates, including input gate $\mathcal{U}_g$ and forget gate $\mathcal{R}_g$ as well as output gate O_g, to successfully manage and learn long-term dependencies. Although LSTM units use more parameters than Gated Recurrent Units (GRUs), they offer a greater level of flexibility in terms of relating complex temporal patterns. The output and internal activity of a LSTM unit are calculated in the following Eqs. (4)-(6):

$$\mathcal{U}_{\mathscr{g}} = \mu\ \left(\mathbb{W}_{\mathcal{U}}\, \mathscr{w}_{\mathscr{s}} + \boldsymbol{Y}_{\mathcal{U}}\mathscr{f}_{\mathscr{s}-1} + \mathscr{p}_{\mathcal{U}}\right) \tag{4}$$

$$\mathcal{R}_{\mathscr{g}} = \mu\ \left(\mathbb{W}_{\mathcal{R}}\, \mathscr{w}_{\mathscr{s}} + \boldsymbol{Y}_{\mathcal{R}}\mathscr{f}_{\mathscr{s}-1} + \mathscr{p}_{\mathcal{R}}\right) \tag{5}$$

$$O_{\mathscr{g}} = \mu\ \left(\mathbb{W}_{O}\, \mathscr{w}_{\mathscr{s}} + \boldsymbol{Y}_{O}\mathscr{f}_{\mathscr{s}-1} + \mathscr{p}_{O}\right) \tag{6}$$

The cell state update C_S is denoted as and the hidden state H_S is equated in (7)- (9),

$$\grave{C}_S = \mathrm{tanH}\left(\mathbb{W}_{C}\, \mathscr{w}_{\mathscr{s}} + \boldsymbol{Y}_{C}\mathscr{f}_{\mathscr{s}-1} + \mathscr{p}_{C}\right) \tag{7}$$

$$C_S = \mathcal{R}_{\mathscr{g}} \odot C_{S-1} + \mathcal{U}_{\mathscr{g}} \odot \grave{C}_S \tag{8}$$

$$H_S = O_{\mathscr{g}} \odot \mathrm{tanH}\ (C_S) \tag{9}$$

Where $\mathscr{w}_{\mathscr{s}}$ represents the input data at time step $\mathscr{s}$, $\mathscr{f}_{\mathscr{s}-1}$ is the hidden state from the previous time step, $\mathbb{W}$ and $\boldsymbol{Y}$ denote weight matrices, $\mathscr{p}$ represents bias terms, μ is the sigmoid activation function, and $\odot$ denotes the Hadamard (element-wise) product.

Attention Mechanism

The attention mechanism creates a context-vector to which various degrees of reputation are attributed to each temporal feature made by the LSTM. This can be done using the basic and context-dependent attention mechanisms (CDAM). These two attention mechanisms hold the power of use individually or in concurrence with each other as per their corresponding attention scores. In this instance, the combined approach is employed, incorporating context-aware attention with the standard LSTM to prioritize the most relevant features. Given that sensor data is highly sensitive to noise, incorporating context from values at other time steps can enhance data classification. Applying a CDAM to both layers of the weighted LSTM allow the algorithm to influence correlations between specific features. The CDAM can be formulated as follows in Eqs. (10)-(12):

$$\mathcal{E}_{\mathscr{s}_i} = \tanh\left(\mathbb{W}_{\mathrm{bc}}\mathscr{f}_{\mathscr{s}_i} + \mathscr{p}_{\mathrm{bc}}\right) \tag{10}$$

$$\gamma_{\mathscr{s}_i} = \frac{exp\mathcal{E}^{\mathcal{S}}_{\mathscr{s}_i}\mathcal{E}_{\lambda}}{\sum_{\mathscr{s}=1}^{\eta} exp\left(\mathcal{E}_{\mathscr{s}_i}, \mathcal{E}_{\lambda}\right)} \tag{11}$$

$$\mathrm{d_i} = \sum\nolimits_{\mathscr{s}=1}^{\eta} \gamma_{\mathscr{s}_i}\mathscr{f}_{\mathscr{s}_i} \tag{12}$$

In this context, $\mathbb{W}_{\mathrm{bc}}$ and $\mathscr{p}_{\mathrm{bc}}$ represent the weighted matrix and biased parameter for the CDAM that need to be learned. Here, $\mathscr{f}_{\mathscr{s}_i}$ denotes the output of the i-th LSTM unit, and $\mathcal{E}_{\mathscr{s}_i}$ are the outputs computed. The score vector $\mathrm{d_i}$ is derived using $\gamma_{\mathscr{s}_i}$. The context information is preserved by $\mathcal{E}_{\lambda}$ which is learned through the network's training process. The context-dependent score vector is produced by summing the comparative weights. While words typically depend on context, EHR data does not always rely on contextual information. In contrast, the attention mechanism which is simplified doesn't require learning the $\mathcal{E}_{\lambda}$ parameter, and it can be determined by using Eqs. (13- (15):

$$\mathcal{E}_{\mathscr{s}_{\mathfrak{k}}} = \tanh\left(\mathbb{W}_{\mathrm{b}}\mathscr{f}_{\mathfrak{k}} + \mathscr{p}_{\mathrm{b}}\right) \tag{13}$$

$$\gamma_{\mathscr{s}_{\mathfrak{k}}} = \frac{exp\mathcal{E}_{\mathscr{s}_{\mathfrak{k}}}}{\sum_{\mathscr{s}=1}^{\eta} exp\left(\mathcal{E}_{\mathscr{s}_{\mathfrak{k}}}\right)} \tag{14}$$

$$\mathrm{d}_{\mathfrak{k}} = \sum\nolimits_{\mathscr{s}=1}^{\eta} \gamma_{\mathscr{s}_{\mathfrak{k}}}\mathscr{f}_{\mathscr{s}_{\mathfrak{k}}} \tag{15}$$

3.3.2 CapsNet for IoT Sensor Data Feature Extraction

The feature extraction layer for IoT sensors is built using capsules. Initially, CapsNet were utilized for image classification. Capsules and the dynamic routing protocol (DRP) are key components of CapsNet. Through a sophisticated interior computational process, a capsule encodes data into a vector. The DRP is introduced as an alternative to max pooling and scalar-output detectors in CNNs. Capsules encode data into vectors, which allows them to capture intricate relationships and hierarchical structures within the sensor data, resulting in a more detailed and informative representation compared to traditional scalar-based approaches. Additionally, the CapsNet incorporates the squash function, designed to normalize the distance of vectors to fall within the range of 0 to 1. The squash function is mathematically defined as follows in Eq. (16):

$$W_{\mathfrak{k}} = SQuash\left(SQ_{\mathfrak{k}}\right) = \frac{\parallel SQ_{\mathfrak{k}} \parallel^2 SQ_{\mathfrak{k}}}{1 + \parallel SQ_{\mathfrak{k}} \parallel^2 \parallel SQ_{\mathfrak{k}} \parallel} \tag{16}$$

Here, the output of the k th capsule is denoted as $W_{\mathfrak{k}}$, and to the vector to be constricted is denoted as $SQ_{\mathfrak{k}}$. The CapsNet is designed to have three layers: a convolutional (CoNv) layer, a primary capsule (PCaps) layer and a digit capsule (DCaps) layer. The value of 256 in the CoNv layer represents the number of CoNv kernels, the dimension dimension of 1×41 represents the size of the kernel and the stride value is 2, which is used to determine the step size. The value of 8 in the PCaps layer corresponds to the number of CoNv kernel groups whereas the number 34 corresponds to the number of kernels in each group. The dimension 1×21 determines the size of each CoNv kernel, and the stride of 2 also specifies the step length. For the DCaps layer, the output parameter of 12 outlines the number of output feature dimensions, and the length parameter of 16 indicates the feature length for each dimension. The CoNv layer is made up of numerous CoNv layers that include ReLU activation functions, described as follows in Eq. (17):

$$Z_{\mathfrak{k}} = ReLU\left(\sum\nolimits_{\mathfrak{i}=1}^{\eta} \mathfrak{q}_{\mathfrak{i}} \mathbb{W}_{\mathfrak{i}\mathfrak{k}} + \not{\mathfrak{b}}_{\mathfrak{k}}\right) \tag{17}$$

Here $\mathfrak{q}_{\mathfrak{i}}$ denotes the $\mathfrak{i}$-th vector, $\mathbb{W}_{\mathfrak{i}\mathfrak{k}}$ is the $\mathfrak{k}$-th weight parameter associated with $\mathfrak{q}_{\mathfrak{i}}$, $\not{\mathfrak{b}}_{\mathfrak{k}}$ represents the biased parameter of the CoNv layer, η is the total number of CoNv kernels, and $Z_{\mathfrak{k}}$ denotes to the output of the CoNv layer. The CoNv layer is responsible for extracting intellectual features for the PCaps layer. The PCaps layer is the initial capsule layer where traditional scalar-output CNNs are replaced by vector-output capsules. In this layer, the Caps-Net utilizes $\mathbb{L}$ sets of CoNv kernels. The PCaps layer processes the output from the CoNv layer to produce vectors that capture spatial features, which can be described as follows in Eq. (18):

$$\mathbb{Q}_{\mathfrak{i}} = SQuash\left(\sum\nolimits_{\mathfrak{k}=1}^{\mathbb{L}} Z_{\mathfrak{k}} \mathbb{W}_{\mathfrak{k}\mathfrak{i}} + \not{\mathfrak{b}}_{\mathfrak{i}}\right) \tag{18}$$

Here $\mathbb{L}$ represents the number of capsules, $\mathbb{W}_{\mathfrak{k}\mathfrak{i}}$ denotes the $\mathfrak{i}$-th weight parameter associated with $Z_{\mathfrak{k}}$, $\not{\mathfrak{b}}_{\mathfrak{k}}$ is the bias parameter for the PCaps layer, and $\mathbb{Q}_{\mathfrak{i}}$ denotes to the output of the PCaps layer. Routing occurs exclusively among consecutive capsule layers. Through an iterative procedure, dynamic routing seeks to establish nonlinear mappings among the PCaps and DCaps layers. The coupling coefficients $\mathcal{C}o_{\mathfrak{i}\mathfrak{k}}$ between these two layers are expressed in Eq. (19):

$$\mathcal{C}o_{\mathfrak{i}\mathfrak{k}} = \frac{exp\left(\not{\mathfrak{a}}_{\mathfrak{i}\mathfrak{k}}\right)}{\sum\nolimits_{\kappa=1}^{\eta} exp\left(\not{\mathfrak{a}}_{\mathfrak{i}\kappa}\right)} \tag{19}$$

Here $\not{\mathfrak{a}}_{\mathfrak{i}\mathfrak{k}}$ represents the initial logit for $\mathcal{C}o_{\mathfrak{i}\mathfrak{k}}$ while $\not{\mathfrak{a}}_{\mathfrak{i}\kappa}$ denotes the dynamic routing weight connecting the $\mathfrak{i}$-th capsule in the PCaps layer with the κ-th capsule in the DCaps layer. The DCaps layer produces the last prediction vector based on the dynamic routing agreement, which can be described as follows Eqs. (20) and (21):

$$SQ_{\mathfrak{k}} = \sum {}_{\mathfrak{i}}\mathcal{C}o_{\mathfrak{i}\mathfrak{k}}\mathfrak{w}_{\mathfrak{k}|\mathfrak{i}} \tag{20}$$

$$\mathfrak{w}_{\mathfrak{k}|\mathfrak{i}} = \mathbb{W}_{\mathfrak{i}\mathfrak{k}}\mathfrak{w}_{\mathfrak{i}} \tag{21}$$

Here $\mathfrak{w}_{\mathfrak{k}|\mathfrak{i}}$ denotes the output of separate capsule in the subsequent layer, $\mathbb{W}_{\mathfrak{i}\mathfrak{k}}$ is the weight parameter, and $SQ_{\mathfrak{k}}$ represents the feature of the input data that has been treated by the DCaps layer. The outputs from ALSTM and CapsNet are fused to generate a comprehensive representation of the patient's health condition, allowing the model to classify individuals into low, moderate, or high-risk categories with high precision. To ensure timely clinical responses, the CURE-HD framework incorporates a Multi-Agent Deep Reinforcement Learning (MA-DRL) scheduler, which prioritizes patient handling based on risk level, resource availability, and waiting time.

3.4 Multi Agent Reinforcement-based Scheduling

The risk classification output is provided to the DRS for HDCP in which the low, moderate, and high-risk patient data are scheduled separately by MA-DRL based on factors such as patient risk score, resource availability, patient wait time, and appointment slot utilization respectively. Figure 4 shows the MA-DRL based task scheduling model.

Based on the schedules, the reports are provided to the MCS which performs heart disease detection separately based on their risk status using three AI models. To design task scheduling algorithm for MA-DRL in this model, we can alleviate problem into multiple steps, concerning on optimizing scheduling according to parameters. Initially, the environment is defined as,

- State $\mathfrak{S}_t$: $\mathfrak{S}_t = \{ \mathrm{Y}_i, \mathrm{a}_j, \mathrm{w}_k, \mathrm{u}_L \}$

Where Y_i is the i-th patient risk score, a_j denotes the availability of resource in medical equipment or personnel j. w_k denotes the waiting time of patient k and u_L is the utilization of appointment dolt L..

- Action Λ_t: $\Lambda_t = \{s_{ij}\}$

Where s_{ij} signifies the scheduling patient i for resource j in time t.

- Reward R_t: $\mathrm{R}_t = \varepsilon_1 \times \mathrm{Y}_i + \varepsilon_2 \times \mathrm{a}_j, \varepsilon_3 \times \mathrm{w}_k, \varepsilon_4 \times \mathrm{u}_L$

Here, $\varepsilon_1, \varepsilon_2, \varepsilon_3$ and ε_4 are the parameters assigned for each factor, summing to 1.

Each agent equivalent to particular group of patients based on risk (low, moderate, high). The agents are trained to determine the optimal policy for scheduling as DRS agent 1- low risk patients, DRS agent 2- moderate risk patients and DRS agent 3- high risk patients. For each agent i, maximize the collective reward R_t over time.

$$\pi_i^* = \arg\max_{\pi_i} \mathbb{E}\left[\sum_{t=0}^{T} \varpi^t \mathrm{R}_{i,t}\right] \tag{22}$$

Form abovementioned Eq. (22), π_i represents the policy for agent i and ϖ is the discount factor of future rewards. Furthermore, Q-learning rule is updated as in Eqs. (23),

$$Q_i(\mathfrak{S}_t, \Lambda_t) = Q_i(\mathfrak{S}_t, \Lambda_t) + \eta \left[\mathrm{R}_{i,t+1} + \varpi \max_{\Lambda'} Q_i(\mathfrak{S}_{t+1}, \Lambda') - Q_i(\mathfrak{S}_t, \Lambda_t)\right] \tag{23}$$

Here, η is the learning rate and Q_i denotes the Q-value function for agent i. In every time step t, observe the current state $\mathfrak{S}_t$, in which each agent chose an action Λ_t corresponding on its policy π_i. Then the patients i are schedule to apposite resource j. Moreover, the reward R_t is evaluated as per instantaneous consequences and the Q-value is updated by updating rule of Q-learning. In decision making and execution, the agent with maximum collective reward R_t in time t might have its scheduling decision conducted. This process is iterating until overall patient data has been scheduled to the multi cloud server system. This configuration allows reshapeless

 https://doi.org/10.1007/s44196-026-01405-w

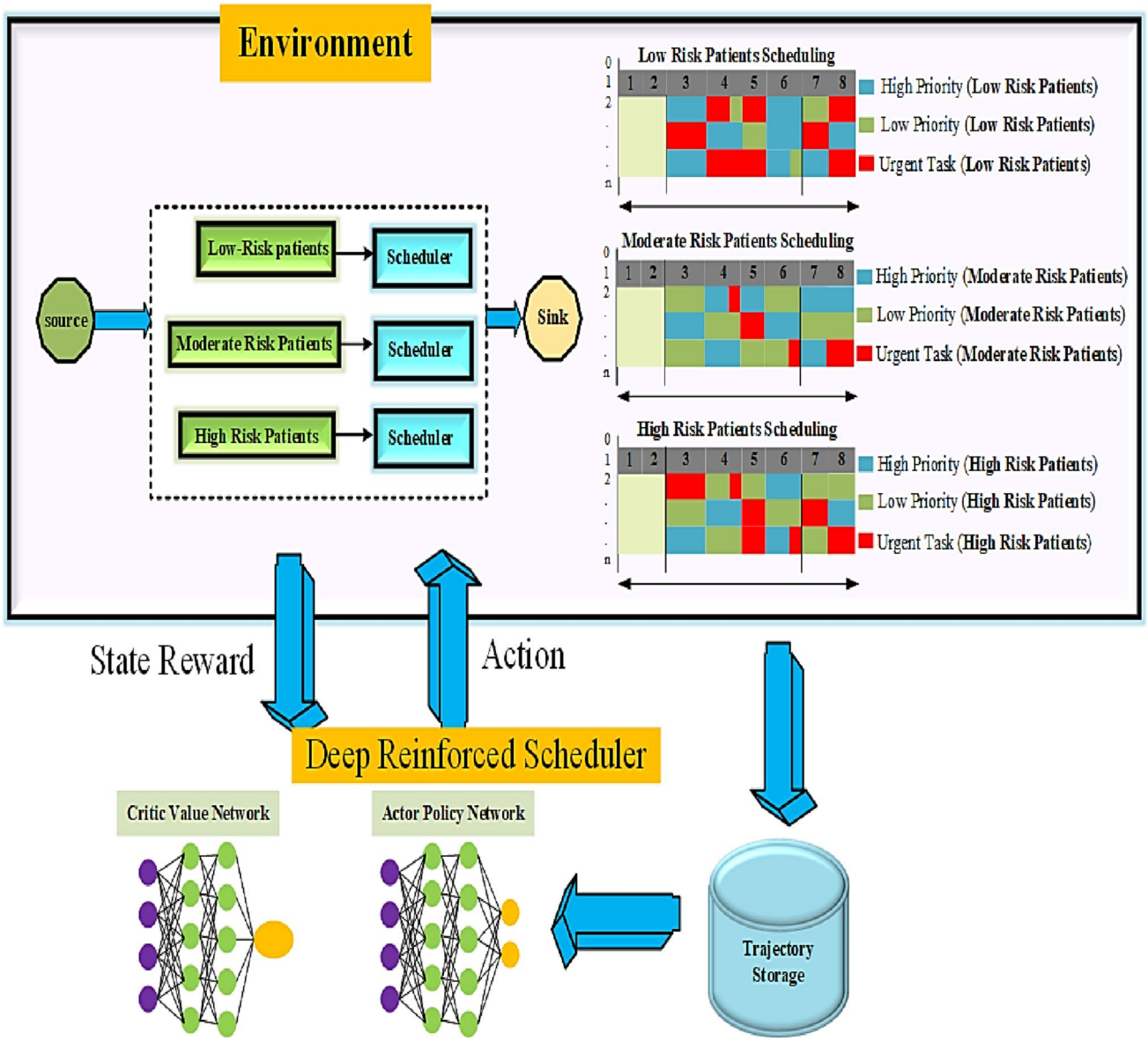


Fig. 4 Structure of MA-DRL based Risk Scheduling Model

and scalable data processing and model training. The model can access more computational resources and collaborative learning opportunities by linking the power of a multi-cloud environment and thus make more accurate and reliable HD predictions.

4 Results and Discussion

In this section, the effectiveness of the proposed CURE-HD model has been analyzed. The practical testing of the proposed model CURE-HD was carried out involving the implementation of cloud-based services and local computing resources and made to be scalable and applicable in real-time. All the simulations were run on an Amazon Cloud environment connected to MATLAB Simulink 2020a to design and validate the model. The local

Table 2 Experimental setup for training the model

Category	Specification / Description
Hardware Configuration	
Processor	AMD Ryzen 5 5600 H @ 3.30 GHz
GPU	NVIDIA GTX 1650 (4 GB VRAM)
RAM	8 GB
Operating System	Windows 11 (64-bit)
Cloud Environment	AWS EC2 (t2.xlarge instance)
Software & Tools	
Programming Environment	MATLAB Simulink 2020a
Deep Learning Frameworks	TensorFlow 2.12, Keras, PyTorch
ML/Utility Libraries	Scikit-learn, NumPy, Pandas
Cloud SDK	AWS SDK for multi-cloud operations
Training Parameters	
Dataset Split	70% Training, 15% Validation, 15% Testing
Batch Size	32
Number of Epochs	100
Optimizer	Adam (learning rate = 0.0001)
Loss Functions	Categorical Cross Entropy, Binary Cross Entropy
Regularization	Dropout (0.3), L2 (0.001)
Routing Iterations (CapsNet)	3

Table 3 Sensor readings based risk level classification

EHR data	Low Risk	Moderate Risk	High Risk
Cholesterol (mg/dL)	<200	200–239	≥ 240
Blood Oxygen (%)	≥ 96	90–95	<90
Respiration Rate (/min)	12–16	17–20	>20 or <10
Blood Glucose (mg/dL) (Fasting)	70–99	100–125 (pre-diabetic)	≥ 126
Heart Rate (bpm)	60–80	81–100	>100 or <50
Blood Pressure (mmHg)	$<120/80$	120–139 / 80–89	$\geq 140/90$

tests were run on a workstation with the processor AMD Ryzen 5 5600 H (3.30 GHz, 6 cores), 8GB RAM, and Windows 11 (64-bit) operating system. Table 2 illustrates the experimental setup for training the model.

Table 3 provides an overview of the typical ranges of values of each sensor and it is divided into low, moderate and high-risk categories according to clinical guidelines. These cutoffs created the labeled data of the classification model and risk stratification. Table 4 indicate the experimental findings of the proposed model.

Table 4 presents a representative data of patients monitored using a combination of EHR and IoT-based physiological sensors. The collected parameters include blood pressure, heart rate, blood glucose, respiration rate, oxygen saturation (SpO_2), and cholesterol levels, which are critical indicators of cardiovascular health. In addition, biosignals such as EEG, EMG, and ECG are collected for deeper physiological analysis. Based on the threshold classification defined in Table 3, each patient is categorized into a low, moderate, or high heart risk category. This consolidated data forms the input to the predictive system and aids in accurate health risk assessment.

4.1 Dataset Details

For evaluating the proposed CURE-HD model, we adopt dataset from UCI repository named Cleveland dataset which can be accessed from the http://archive.ics.uci.edu/ml/datasets.php. The data presented in the dataset had own characteristics and instances, that composed of 78 attributes with 305 data records. In that, we adopt only 14 attributes for testing and training the proposed model. To be clearer, the complete dataset was utilized to train the designed model whereas the acquired sensor data are utilized for model testing. From the training data the

https://doi.org/10.1007/s44196-026-01405-w

Table 4 Experimental results of the proposed model

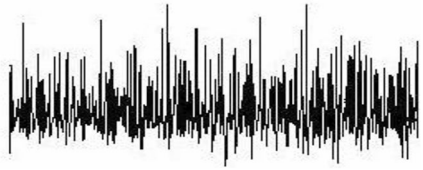
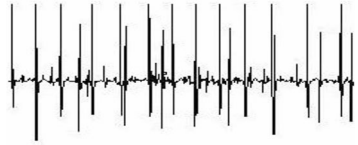
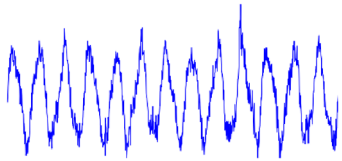
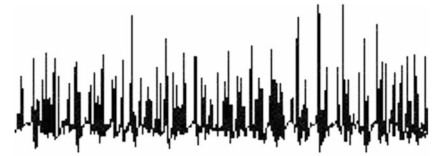
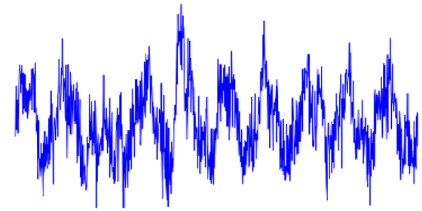
Patient_id	Input Data			Result
	Medical data	IoT Sensor data		
001	Name: Akash Sex: Male Age: 24 EHR data: BP:115/75 Heart Rate: 72 Glucose: 90 Resp Rate: 14 SpO_2 (%): 98 Cholesterol:180	EEG signal		Low Risk
		EMG signal		
		ECG signal		
002	Name: Sharon Sex: Female Age: 32 EHR data: BP:130/85 Heart Rate: 88 Glucose: 110 Resp Rate: 18 SpO_2 (%): 93 Cholesterol:220	EEG signal		Moderate risk
		EMG signal		
		ECG signal		

Table 4 (continued)

Patient_id	Input Data			Result
	Medical data	IoT Sensor data		
003	Name: Arun Sex: Male Age: 42 EHR data: BP:155/95 Heart Rate: 105 Glucose: 140 Resp Rate: 22 SpO_2 (%): 87 Cholesterol:260	EEG signal		High risk
		EMG signal		
		ECG signal		

features are extracted by the testing data. We have also utilized echocardiogram image dataset for examining the proposed heart disease model. For retrieving the ECG images, we utilize UCI repository that composed of 70 abnormal ECG image with 40 subjects, and 70 normal ECG image with 40 subjects. The training and testing process is similar to the above dataset.

4.2 Performance Metrics

In this section, we utilized performance metrics like F1-score, recall, precision, and accuracy respectively for validating the proposed model. The validation metrics equations are shown from Eqs. (24)-(28),

$$\mathrm{Acc} = \frac{\mathrm{TrPV} + \mathrm{TrPN}}{\mathrm{TrPV} + \mathrm{FaLP} + \mathrm{TrPN} + \mathrm{FaLN}} \tag{24}$$

$$\mathrm{Pre} = \frac{\mathrm{TrPV}}{\mathrm{TrPV} + \mathrm{FaLP}} \tag{25}$$

$$\mathrm{Recc} = \frac{\mathrm{TrPV}}{\mathrm{TrPV} + \mathrm{FaLN}} \tag{26}$$

$$\mathrm{Spe} = \frac{\mathrm{TrPN}}{\mathrm{TrPN} + \mathrm{FaLP}} \tag{27}$$

$$\mathrm{F1S} = \frac{2\mathrm{TrPV}}{2\mathrm{TrPV} + \mathrm{FaLP} + \mathrm{TrPN}} \tag{28}$$

From the above equations, TrPV, TrPN, FaLP, and FaLN denotes the true positive, true negative, false positive, and false negative values respectively in normal and abnormal classes. In order to examine the performance of the dataset abnormal and normal classes, we utilized proposed CURE-HD model which processes the dataset (i.e. EHR and ECG), and collected sensor data respectively with the mentioned performance metrices.

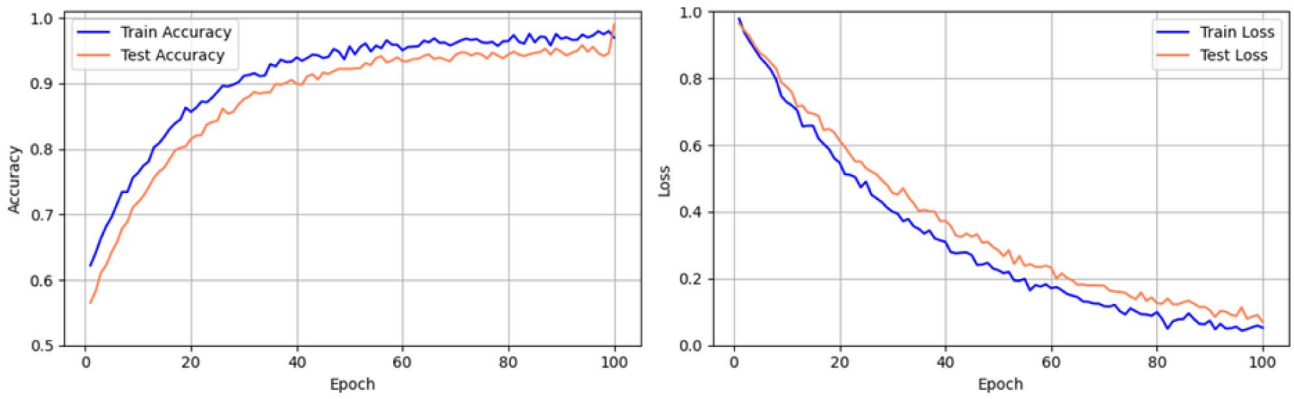
4.3 Performance Analysis

The performance of the proposed CURE-HD model has been assessed using various quantitative metrics, including accuracy, specificity, sensitivity, and confusion matrix analysis. in this section.

Figure 5 (a) represents the accuracy and 5(b) represents the loss graph. The accuracy and loss graph of the developed ALSTM-CapsNet approach is shown in Fig. 5. The accuracy graph illustrates the evolution of the method's efficacy over time on training and testing datasets. As epochs increase, both the training and testing accuracies rise steadily, eventually plateauing. The training accuracy consistently stays higher than the testing accuracy, indicating that the model is learning effectively, but there might be slight overfitting as the gap between the two increases toward the later epochs. In contrast, the loss graph shows a decreasing trend for both training and testing losses, reflecting the model's improvement in minimizing error as it learns, with both loss curves converging toward low values, which demonstrates the successful training of the suggested ALSTM-CapsNet approach.

The Receiver Operating Characteristic (ROC) curve depicted in Fig. 6 is employed to assess the classification efficacy of the suggested approach. By plotting the TP rate against the FP rate, the ROC curve connects the trade-off of sensitivity (recall) and specificity at different threshold levels. The larger the AUC value, the greater the classification performance, which indicates the accuracy level of the model.

Figure 7 reveals the confusion matrix of the suggested CURE-HD model to classify patients as High Risk, Moderate Risk and Low Risk. The model presents extremely high classification accuracy in all classes with the



(a) (b)

Fig. 5 Accuracy and loss graph

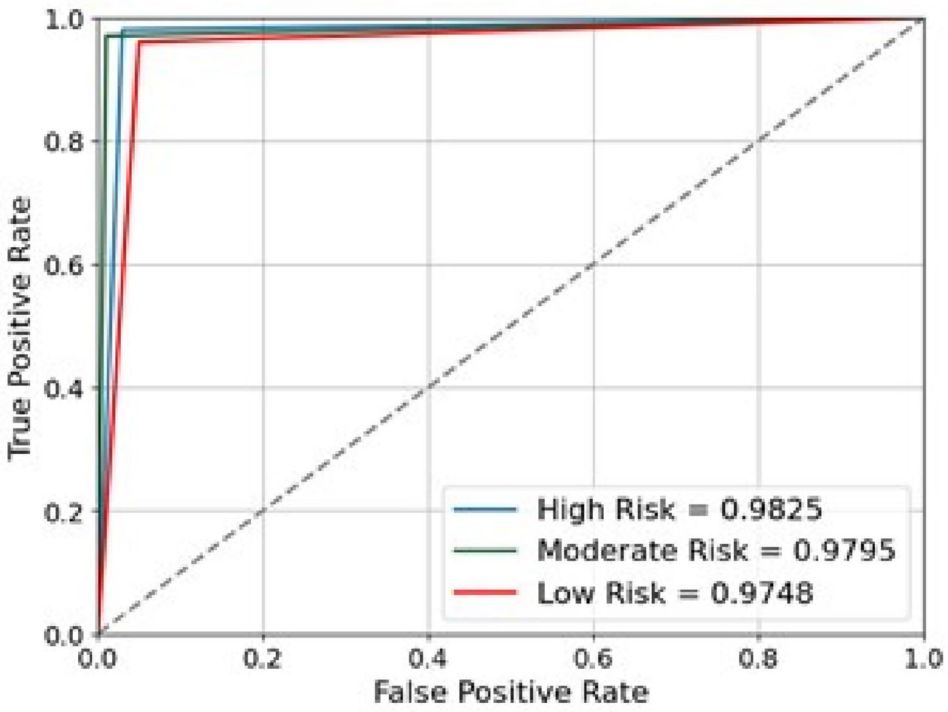


Fig. 6 ROC graph of proposed CURE-HD model

correct prediction rates of 98.25(High Risk), 97.95 (Moderate Risk), and 97.48 (Low Risk). The misclassification error measures are also small with less than 1.52% between any two classes. The findings suggest that CURE-HD is able to accurately differentiate between the different levels of risk, which guarantees the high sensitivity and specificity in clinically useful situations.

Table 5 provides the performance of the proposed CURE-HD on high risk, moderate risk, and low risk cases. The proposed CURE-HD was trained and tested with performance metrics such as accuracy, precision, recall, specificity, and F1-score respectively. With the collected sensor data, the proposed model acquires 98.59% of accuracy, 96.67% of precision, 99.01% of recall, 97.91% of specificity, and 98.22% of F1-score for the high risk classes. With the UCI dataset (EHR and ECG), the model acquires accuracy of 97.42%, 97.59% of precision, 99.94% of recall, 98.63% of specificity, and 98.89% of F1-score for the high risk classes. For the moderate risk class with collected sensor data, the accuracy, precision, recall, specificity, F1score acquired are 98.72%, 97.12%,

 https://doi.org/10.1007/s44196-026-01405-w

Fig. 7 Confusion matrix of proposed CURE-HD model

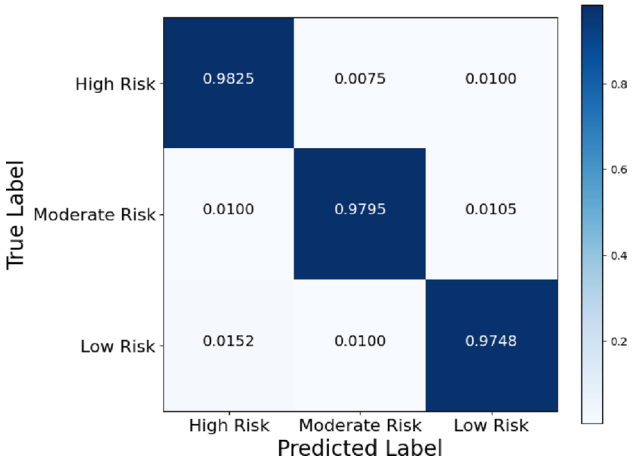


Table 5 Performance analysis of the proposed CURE-HD for high risk, moderate risk and low risk cases

Class	Data	Acc	Pre	Rec	Spe	F1S
High risk	Sensor Data	98.59%	96.67%	99.01%	97.91%	98.22%
	UCI Dataset (EHR and ECG)	97.42%	97.59%	99.94%	98.63%	98.89%
Moderate risk	Sensor Data	98.72%	97.12%	98.14%	98.25%	98.54%
	UCI Dataset (EHR and ECG)	98.11%	97.25%	99.41%	98.51%	98.81%
Low risk	Sensor Data	99.64%	97.83%	99.85%	96.36%	99.26%
	UCI Dataset (EHR and ECG)	97.94%	96.21%	98.15%	95.57%	96.34%

99.85%, 98.14%, 98.25% respectively. For the low risk classes with EHR and ECG data, the accuracy, precision, recall, specificity, F1-score are 97.94%, 96.21%, 98.15%, 95.57%, and 96.34%. For the low risk class with collected sensor data, the accuracy, precision, recall, specificity, F1score acquired are 99.64%, 97.83%, 99.85%, 96.36%, 99.26% respectively. For the low risk classes with EHR and ECG data, the accuracy, precision, recall, specificity, F1-score are 97.94%, 96.21%, 98.15%, 95.57%, and 96.34% respectively

4.4 Comparative Analysis

In this section, the proposed CURE-HD technique has been compared with the baseline methods, such as FRIEND [16], DEEP-CARDIO [18], and Heart Sense [22] models, in terms of performance metrics such as specificity, precision, recall, accuracy, F1-score, and RMSE.

Figure 8 represents the comparison of the developed CURE-HD method with the state-of-art (SOTA) techniques such as FRIEND [16], DEEP-CARDIO [18], and Heart Sense [22] in terms of parameters involving recall, f1-score, precision, and accuracy. The comparison shows that the accuracy of the novel CURE-HD model is 99.2%, whereas the accuracy of the current methods, including FRIEND, DEEP-CARDIO, and Heart Sense models, is 94.27%, 97.35%, and 98.89%, respectively. From the results, the proposed CURE-HD increases the overall accuracy by 11.38%, 10.92%, and 5.51% for FRIEND, DEEP-CARDIO, and Heart Sense respectively. The proposed CURE-HD surpasses the FRIEND, DEEP-CARDIO, and Heart Sense models by enhancing the efficiency and accuracy in predicting risk.

Figure 9 (a) and 9 (b) compares the specificity and sensitivity of the proposed CURE-HD technique with the existing techniques such as FRIEND, DEEP-CARDIO, and Heart Sense models. In both specificity and sensitivity, the proposed framework CURE-HD maintains the highest value, showing that it is best capable of correctly identifying high risk, medium risk and low risk. CURE-HD maintains specificity greater than 97.5% and sensitivity approaching 100% for all sample sizes, indicating very high reliability and robustness. While the other methods seem to fluctuate in their performances with the different sample sizes, the proposed CURE-HD

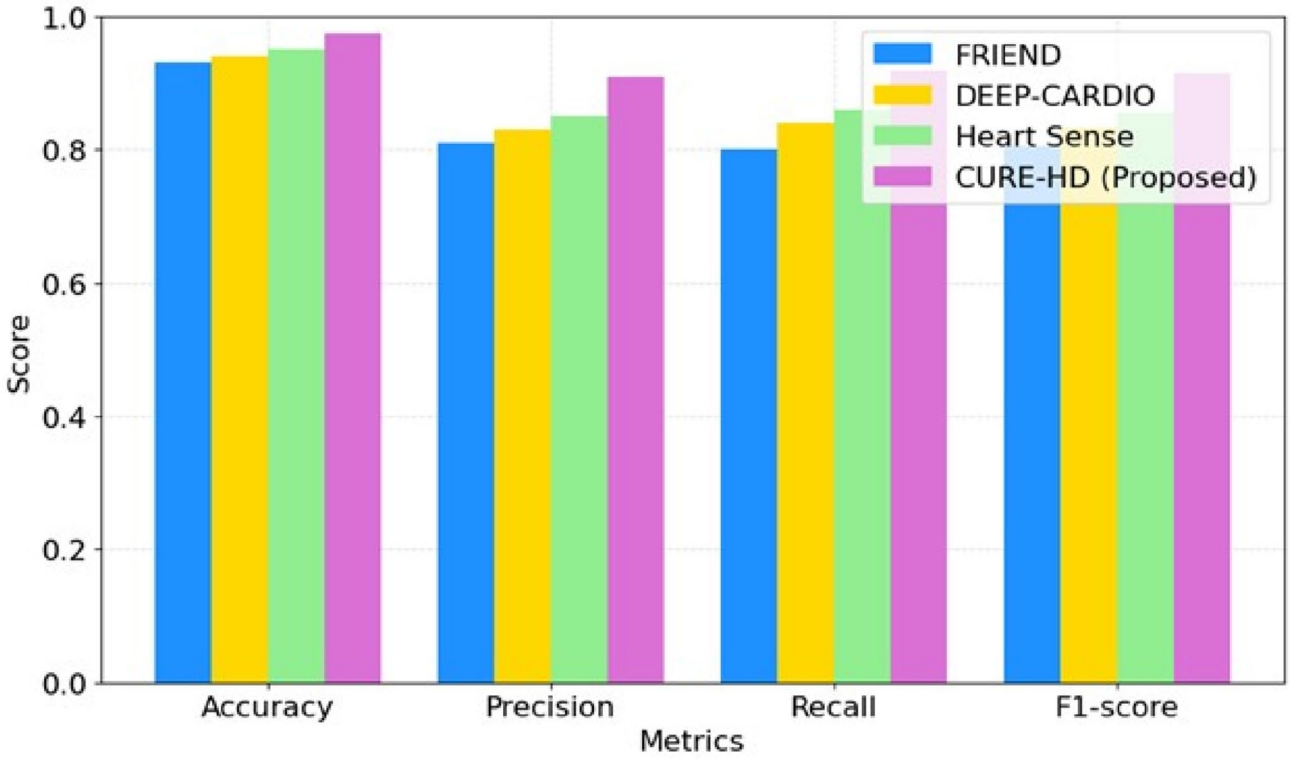


Fig. 8 Performance comparison of proposed CURE-HD model with the existing techniques

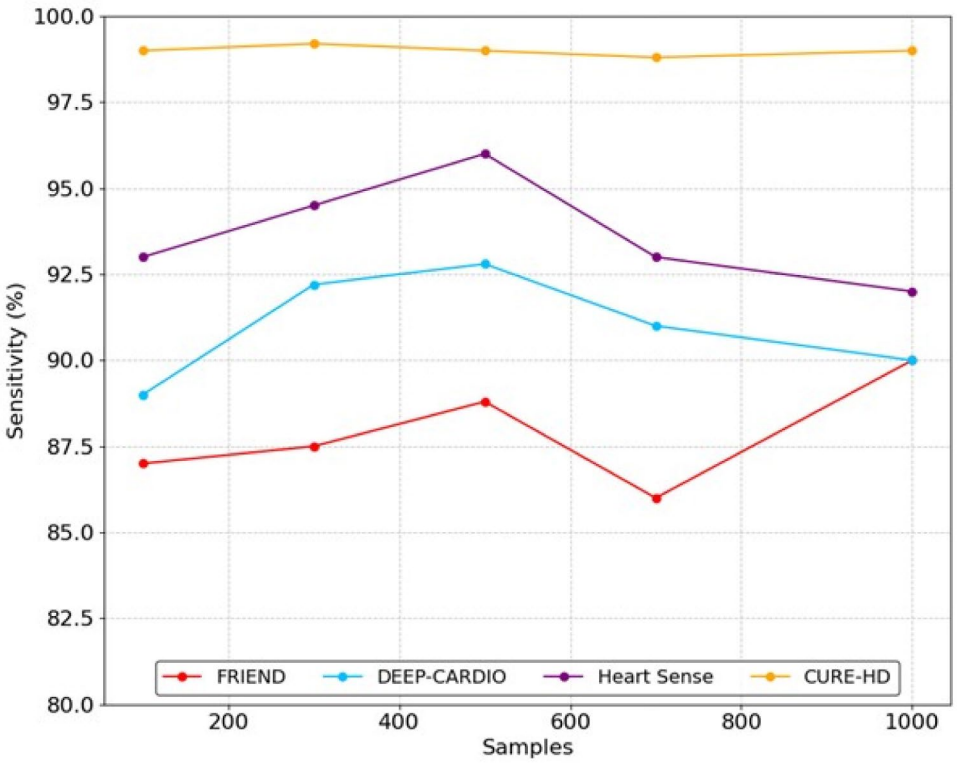
maintains stable results, demonstrating its efficiency with different volumes of data without much degradation in detection accuracy.

Table 6 demonstrates the ablation experiment, which was conducted to determine the role of each significant element in the CURE-HD framework. The performance of the full CURE-HD model is highest with an accuracy of 99.01, precision of 97.12, recall of 98.40 as well as F1-score of 97.75, which proves the effectiveness of the complete architecture. The removal of the MA-DRL scheduling module reduces the accuracy to 96.84, which shows that intelligent task scheduling is an essential component of enhancing diagnostic efficiency. The remaining exclusion of the CapsNet module further lowers accuracy to 95.72, which shows that sensor-based capsule features extraction is critical to absorb physiological differences. Equally, accuracy decreases to 94.91 with the deletion of the ALSTM module, indicating that temporal information in EHR data is critical in classification of risk. All in all, the results of the ablation indicate that every component plays a significant role in the strong performance of the CURE-HD model.

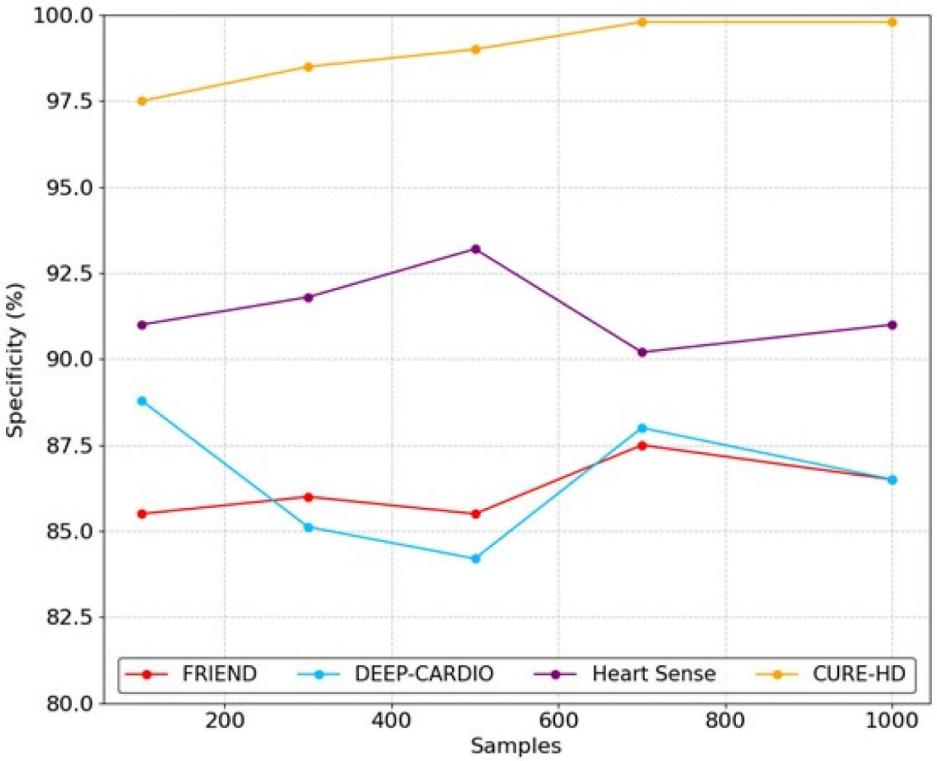
4.5 Discussion

The results indicate that the proposed CURE-HD framework achieves highly reliable and accurate predictions of heart disease risk by integrating electronic health record (EHR) data, IoT sensor signals, and ECG images. Figures 5(a) and 5(b) illustrate the training behavior of the ALSTM-CapsNet model, which converges steadily, as evidenced by the consistent changes in accuracy and loss. The classification performance, summarized in Table 5, shows that the model consistently aligns with risk levels, achieving an impressive overall accuracy of up to 99.01%. Additionally, it demonstrates a high recall for high-risk cases, reaching 99.94% based on UCI data, ensuring that critical cases are effectively identified. Figure 6 presents the ROC curve, which has near-perfect AUC values, while Fig. 7 indicates a minimal amount of misclassification. This suggests that the ROC effectively

 https://doi.org/10.1007/s44196-026-01405-w



(a)



(b)

Fig. 9 Comparison of (**a**) Specificity and (**b**) Sensitivity of proposed CURE-HD with the existing techniques

Table 6 Ablation Study of the Proposed CURE-HD Model

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Full CURE-HD (Proposed)	99.01	97.12	98.40	97.75
CURE-HD w/o MA-DRL	96.84	95.21	97.32	96.25
CURE-HD w/o CapsNet	95.72	94.58	96.10	95.33
CURE-HD w/o ALSTM	94.91	93.77	95.42	94.58

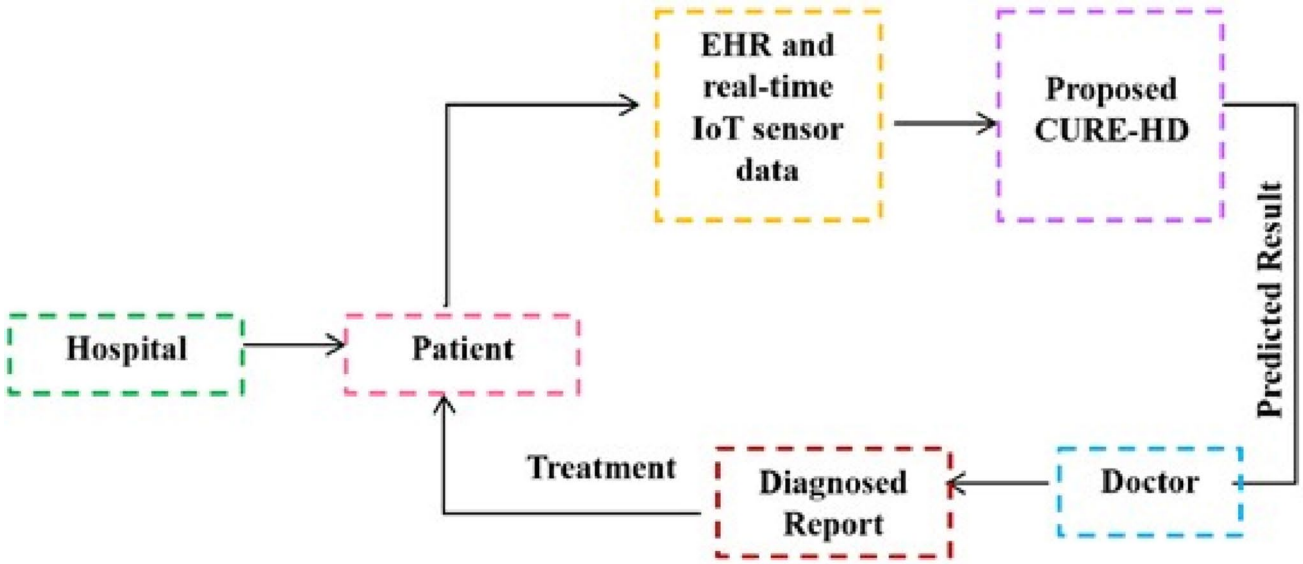


Fig. 10 Real time clinical setting of the CURE-HD

distinguishes between low-, moderate-, and high-risk groups. Ablation studies reveal that removing the attention mechanism, CapsNet, or MA-DRL components decreases accuracy by 1.5–3.2%. This finding underscores the importance of these components in enhancing feature extraction and optimizing scheduling. The MA-DRL scheduling process diagram in Fig. 4 highlights how patient prioritization is determined based on risk level, wait time, and available resources. Comparative analyses illustrated in Figs. 8 and 9 reveal that CURE-HD outperforms three existing methods FRIEND, DEEP-CARDIO, and Heart Sense by enhancing accuracy by 11.38%, 10.92%, and 5.51%, respectively. These findings demonstrate that the hybrid multi-modal architecture and integrated cloud-based scheduling offer significant performance advantages.

4.6 Clinical Settings

This section illustrates a clinical use of the CURE-HD framework in detecting the risks of heart diseases based on patient health data stored via the IoT devices and electronic medical systems. Figure 10 shows the real-time clinical workflow of CURE-HD. It starts with the patient coming to the hospital or being monitored at home with the help of the IoT-based sensors that provide physiological indicators, including ECG, EMG, heart rate, blood pressure, respiration rate, oxygen saturation, cholesterol, and glucose, and Electronic Health Record (EHR)-related data. These are then fed to the Distributed Edge Server (DES) and sent to the ALSTM-CapsNet model to classify the patients effectively as low, moderate and high risk. The Multi-Agent Deep Reinforcement Learning (MA-DRL) scheduler then processes prediction results and helps the cardiologists by ranking the patients according to the level of risk and clinical urgency so that patients can be diagnosed in time and the necessary resources can be properly allocated. This simplified pipeline is not only capable of empowering clinicians, but also making diagnostic delays shorter, and computational load lower, by offering real-time, interpretable, and high-precision risk scores. Moreover, the CURE-HD framework can be easily interconnected with the hospital information systems,

cloud infrastructure, and remote monitoring systems, which would contribute to its increased real-life usability in the contemporary clinical setting.

5 Conclusion

This paper presented a novel multi-Cloud-based Unified Risk Evaluation for Heart Disease (CURE-HD) for detecting heart diseases. The proposed model utilized medical IoT devices for collecting data from the patients and pre-processed data and then forwarded to the deep learning based three level risk classification (i.e. low, moderate, and high), and multi agent reinforcement learning based patient scheduling. Based on the schedule, the doctors will be alerted. For the experimental analysis, the proposed CURE-HD was evaluated using accuracy, specificity, sensitivity, precision, and recall. The proposed CURE-HD model achieves an overall accuracy of 99.01% for heart disease detection. From the results, the proposed CURE-HD increases the overall accuracy by 11.38%, 10.92%, and 5.51% for FRIEND, DEEP-CARDIO, and Heart Sense respectively. Future work could focus on enhancing the scalability of the CURE-HD by optimizing the DL models for real-time patient scheduling could further reduce time complexity.

Acknowledgements The author would like to express his heartfelt gratitude to the supervisor for his guidance and unwavering support during this research for his guidance and support.

Author Contributions The authors confirm contribution to the paper as follows: Study conception and design: Anandhi S.V, Mohana Suganthi; Data collection: Ramakantha Reddy B, Michael Priya E, Mohana Suganthi; Analysis and interpretation of results: Ramakantha Reddy B, Michael Priya E, Mohana Suganthi; Draft manuscript preparation: Ramakantha Reddy B, Michael Priya E. All authors reviewed the results and approved the final version of the manuscript.

Funding No Financial support.

Data Availability The datasets generated and/or analysed during the current study are available from the corresponding author upon reasonable request.

Declarations

Competing Interests The authors declare no competing interests.

Ethical Approval My research guide reviewed and ethically approved this manuscript for publishing in this Journal.

Human and Animal Rights This article does not contain any studies with human or animal subjects performed by any of the authors.

Informed Consent I certify that I have explained the nature and purpose of this study to the above-named individual, and I have discussed the potential benefits of this study participation. The questions the individual had about this study have been answered, and we will always be available to address future questions.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

1. Keikhosrokiani, P., Kamaruddin, N.S.A.B.: IoT-Based in-hospital-in-home heart disease remote monitoring system with machine learning features for decision making. In Connected e-Health: Integrated IoT and cloud computing, 349–369 (2022)
2. Pandey, V., Lilhore, U.K., Walia, R., Alroobaea, R., Alsafyani, M., Baqasah, A.M., Algarni, S.: Enhancing heart disease classification with M2MASC and CNN-BiLSTM integration for improved accuracy. Sci. Rep. **14**(1), 24221 (2024)
3. Yenurkar, G., Mal, S., Nyangaresi, V.O., Kamble, S., Damahe, L., Bankar, N.: Revolutionizing chronic heart disease management: The role of IoT-based ambulatory blood pressure monitoring system. Diagnostics. **14**(12), 1297 (2024)
4. Almujally, N.A., Aljrees, T., Saidani, O., Umer, M., Faheem, Z.B., Abuzinadah, N., Ashraf, I.: Monitoring acute heart failure patients using internet-of-things-based smart monitoring system. Sensors. **23**(10), 4580 (2023)
5. Pandey, V., Lilhore, U.K., Walia, R.: A systematic review on cardiovascular disease detection and classification. Biomed. Signal Process. Control. **102**, 107329 (2025)
6. Khekare, G., Yenurkar, G., Turukmane, A.V., Ameta, G.K., Sharma, P., Phulre, A.K.: Artificial intelligence algorithms for better decision-making. In: Multi-criteria decision-making and optimum design with machine learning, pp. 252–262. CRC (2025)
7. Pandey, V., Lilhore, U.K. & Walia, R. Enhanced multiclass heart disease classification through advanced signal processing with modified mixed attention mechanism-based deep BiLSTM. Multimed Tools Appl **84**, 38893–38910 (2025)
8. Yenurkar, G.K., Mal, S., Wakulkar, A., Umbarkar, K., Bhat, A., Bhasharkar, A., Pathade, A.: Future prediction for precautionary measures associated with heart-related issues based on IoT prototype. Multimedia Tools Appl. **83**(23), 63723–63753 (2024)
9. Pandey, V., Lilhore, U.K., Walia, R.: Advanced Heart Disease Prediction Through Spatial and Temporal Feature Learning with SCN-Deep BiLSTM. Int. J. Comput. Intell. Syst. **18**(1), 25 (2025)
10. Tsao, Y.C., Cheng, F.J., Li, Y.H., Liao, L.D.: An IoT-Based Smart System with an MQTT Broker for Individual Patient Vital Sign Monitoring in Potential Emergency or Prehospital Applications. Emerg. Med. Int. **2022**(1), 7245650 (2022)
11. Makade, J., Bankar, N., Kumar, A., Bandre, G., Yenurkar, G.: Artificial intelligence in health care: A review of uses, challenges and potential uses. In AIP Conference Proceedings, 3188 (1), 080032 (2024)
12. Madumidha Subramanian, S., Paramasivam, A.W.A.R.E.-A.G.R.I.: Real Time Sensors-based Adaptive Weather Forecasting for Agriculture. Int. J. Comput. Eng. Optim. **01**(02), 69–74 (2023)
13. Kapse, S., Yenurkar, G., Nyangaresi, V.O., Balpande, G., Kale, S., Jadhav, M., Lawankar, S., Jaunjale, V.: MediServe: An IoT-Enhanced Deep Learning Framework for Personalized Medication Management for Elderly Care. Computers Mater. Continua, **83**(1), 935-976 (2025)
14. Hemamalini, V., Anand, L., Nachiyappan, S., Geeitha, S., Motupalli, V.R., Kumar, R., Ahilan, A., Rajesh, M.: Integrating bio medical sensors in detecting hidden signatures of COVID-19 with Artificial intelligence. Measurement. **194**, 111054 (2022)
15. Kolli, S., Patro, P., Sharma, R. and Sharma, A. (2024). Classification and Diagnosis of Heart Diseases Using Fuzzy Logic Based on IoT. In Advances in Fuzzy-Based Internet of Medical Things (IoMT) (eds S.P. Yadav, S. Yadav, P.R. Chelliah and V.H.C. Albuquerque). https://doi.org/10.1002/9781394242252.ch10
16. Pati, A., Parhi, M., Alnabhan, M., Pattanayak, B.K., Habboush, A.K., Nawayseh, A.: An IoT-fog-cloud integrated framework for real-time remote cardiovascular disease diagnosis. Inf. **10**(1), 21 (2023)
17. Raheja, N., Manocha, A.K.: An IoT enabled secured clinical health care framework for diagnosis of heart diseases. Biomed. Signal Process. Control. **80**, 104368 (2023)
18. Yashudas, A., Gupta, D., Prashant, G.C., Dua, A., AlQahtani, D., Reddy, A.S.K.: DEEP-CARDIO: Recommendation System for Cardiovascular Disease Prediction using IOT Network. IEEE Sens. J. **24**(9), 14539–14547 (2024)
19. Durga, S., Daniel, E., Andrew, J., Bhat, R., SmartCardio: Advancing cardiac risk prediction through Internet of things and edge cloud intelligence. IET Wirel. Sens. Syst. **14**(6), 348–362 (2024)
20. Ramesh, B., Lakshmanna, K.: A novel early detection and prevention of coronary heart disease framework using hybrid deep learning model and neural fuzzy inference system. IEEe Access. **12**, 26683–26695 (2024)
21. Mandava, M.: MDensNet201-IDRSRNet: Efficient cardiovascular disease prediction system using hybrid deep learning. Biomed. Signal Process. Control. **93**, 106147 (2024)
22. Ghilzai, R.J., Bibi, U., Umer, H.G.A., Rubab, S.: Heart Sense. A novel IoT integrated Deep Learning Based ECG Image Analysis for Enhanced Heart Disease Prediction. Int. J. Innov. Sci. Technol. **7**(1), 336–357 (2025)
23. Sekar, J., Aruchamy: A novel approach for heart disease prediction using hybridized AITH2O algorithm and SANFIS classifier. Network: Comput. Neural Syst. **36**(1), 109–147 (2025)
24. Chichani, Y.S., Kasar, S.L.: An efficient IoT enabled heart disease prediction model using Finch hunt optimization modified BiLSTM classifier. Biomed. Signal Process. Control. **100**, 107170 (2025)
25. Tarek, Z., Alhussan, A.A., Khafaga, D.S., El-Kenawy, E.S.M., Elshewey, A.M.: A snake optimization algorithm-based feature selection framework for rapid detection of cardiovascular disease in its early stages. Biomed. Signal Process. Control. **102**, 107417 (2025)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

S. V. Anandhi[1] · Mohana Suganthi[2] · B. Ramakantha Reddy[3] · S. Priya[4]

✉ S. Priya
priyasugam@gmail.com

[1] Department of Artificial Intelligence and Data Science, RAMCO Institute of Technology, Rajapalayam 626117, Tamil Nadu, India

[2] Department of Computer Science and Engineering, Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai 600062, Tamil Nadu, India

[3] Department of Computer Science and Engineering (AI ML), Sri Venkateswara College of Engineering, JNTU Anantapur, Tirupati, India

[4] Department of Computer Science and Engineering (CS) School of Computing, Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Avadi, Chennai 600 062, Tamil Nadu, India